TOPOSCOUT

GNSS SURVEY PLANNING SYSTEM

SITE INTELLIGENCE REPORT

REPORT SG-89D6FJ
Generated July 13, 2026
Data current as of report date

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
38.3459°N 122.3047°W WGS84	**66 ft** NAVD88 · MSL	**CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)**	**Zone X** 06055C0510FF	**13.0° E** WMM 2026

APN **038050009000** | ZONING **AP** | ACRES **0.481 ac**

BEST GNSS WINDOW **Mon, Jul 13 · 6:00 AM – 10:00 AM** | **PDOP 0.99 · 34 sv** | NEAREST CORS **P261 · 14.1 mi ✓**

REPORT CONTENTS

SCAN FOR KMZ
Site file · Google Earth
↓ Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St, Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

SITE & SIGNAL REPORT

COORDINATES
38.3459, -122.3047
WGS84 / NAD83

ELEVATION
66 ft
Above MSL (approx)

MAGNETIC DECLINATION
13.0° E
Compass reads 13.0° too far W

FEMA FLOOD ZONE
X
AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX
4.7

AVG KP
2.4

SOLAR FLUX F10.7
138 sfu

TEC (VTEC) ⚠ DEGRADED
19.0 TECU
-3.8 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location — orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

🔥 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change — verify before fieldwork

🛰 GPS CONSTELLATION HEALTH

⚠ 1 Satellite Currently Unhealthy — PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window

PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Monday, Jul 13

Overcast | 💨 9 mph | **GO**

High 93°F | Low 59°F | Wind max 9 mph | Gusts 12 mph | Precip 0.00" | Sunrise 5:56 AM | Sunset 8:33 PM

🌘 Waning Crescent 2%

GNSS: Avg PDOP **0.99** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP — HOURLY

	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL

⛅ **GO — GOOD CONDITIONS**

Partly cloudy | Wind: 9.4 mph (gusts 11.6 mph) | Precip: 0" | Temp: 59°–93.2°F

🚁 DRONE WIND ANALYSIS

✅ **GO** Peak: 9.4/11.6 mph S

☀ Sunrise: **5:56 AM** ☾ Sunset: **8:33 PM** fly window clipped to daylight

✅ Best flight window: **6:00 AM – 9:00 PM (15 hrs)**

🔄 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph ⚡=turbulence	DIR
5 AM	2/4 mph	SE
6 AM	0/3 mph	W
7 AM	3/5 mph	E
8 AM	2/2 mph	SSW
9 AM	2/3 mph	SE
10 AM	4/5 mph	SSE
11 AM	5/5 mph	S
12 PM	5/6 mph	S
1 PM	5/5 mph	S
2 PM	9/10 mph	S
3 PM	7/8 mph	S
4 PM	8/9 mph	SSW
5 PM	9/11 mph	SSW
6 PM	9/11 mph	SSW
7 PM	9/12 mph	S
8 PM	8/11 mph	S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS A** ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
✈ TFR STATUS ✅ **NO TFRs WITHIN 30 MI** Check B4UFLY ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

AI FIELD ANALYSIS — CLAUDE HAIKU 4.5

FIELD PLANNING REPORT

GNSS Survey Operations — Napa County

Report Date: Monday, July 13, 2026 | Survey Start: Monday, July 13, 2026

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VERDICT

GO — Favorable PDOP geometry (0.94–0.99), acceptable weather, and moderate ionospheric activity present no barrier to operations. Survey may proceed with standard static protocols.

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BEST SURVEY WINDOW

Monday, July 13, 2026 | 06:00 AM – 10:00 AM (4-hour window)

Rationale: PDOP 0.94 (excellent), 34 visible satellites (multi-constellation), clear geometry across window. Peak constellation strength occurs during this interval; recommend prioritizing static sessions and base-station observations in this timeframe.

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BEST 4-HOUR STATIC GNSS WINDOW

Monday, July 13, 2026 | 06:00 AM – 10:00 AM

Occupation Time Recommendation: Minimum 45–60 minutes per point (standard baseline < 20 km). Extend to 60–90 minutes if baseline exceeds 50 km or if ionospheric TEC uncertainty (currently –3.8 TECU anomaly vs. 30-day median) is a concern. Multi-constellation lock (GPS/GLONASS/Galileo/BeiDou) will mitigate moderate geomagnetic activity (Kp avg 2.4, peak 4.7).

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DRONE OPERATIONS ASSESSMENT

APPROVED FOR VFR OPERATIONS. Overcast conditions (93°F high, 59°F low) present no thermal or visibility constraint. Wind max 9 mph is well within limits for small-UAS operations. No precipitation forecast. Recommend standard pre-flight checks and line-of-sight compliance. No air-quality barrier (AQI 45 = Good).

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FIELD SAFETY WARNINGS

SATELLITE PRN 11 (GPS) — Currently unhealthy (UNUSUFN). Receivers will auto-exclude this satellite; PDOP calculations already reflect this loss. No manual intervention required.

No active wildfires within reporting range. All-clear. Air quality acceptable. Standard field PPE sufficient.

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SUMMARY

Survey conditions on Monday, July 13, 2026, are favorable. Excellent PDOP (0.94–0.99) during the 06:00–10:00 AM window, combined with acceptable weather (overcast, 9 mph wind, no precipitation), permits confident execution of static GNSS observations. Moderate ionospheric activity (Kp avg 2.4) is routine; standard occupation times (45–90 minutes per point, multi-constellation lock) will ensure centimeter-level accuracy. One GPS satellite (PRN 11) is unhealthy but auto-excluded by receivers. No operational or safety impediments identified.

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COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

ASSESSOR / TAX OFFICE

Napa County Recorder-Clerk	Napa County Assessor
1127 First St, Napa, CA 94559	1127 First St, Napa, CA 94559

⚡ Nearby Control Points · 38.3459, -122.3047

CORS STATIONS — WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
NGS ↗	**P261**	HUNTERHILLCN2004	14.1 mi
NGS ↗	**P198**	PETALUMAIRCN2004	17.5 mi
NGS ↗	**OHLN**	OHLONE PARK	23.5 mi
NGS ↗	**P196**	MEACHUMLFLCN2006	24.0 mi
NGS ↗	**CASR**	SANTA ROSA CA	24.9 mi
NGS ↗	**P267**	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS — WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗	**JT9010**	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9018**	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9009**	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9017**	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9034**	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
Datasheet ↗	**JT9012**	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
Datasheet ↗	**JT9043**	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X — AREA OF MINIMAL FLOOD HAZARD

Panel: **06055C0510FF**
Effective: Sep 29, 2010
View FEMA Map ↗

DATA COLLECTOR IMPORT — NGS BENCHMARKS

↓ Download .txt File

```
NGS BENCHMARK CONTROL POINTS — Copy / Paste to Data Collector
Datum: NAD83  |  Vertical: NAVD88  |  Source: NGS NDE API

PID      NAME                    LAT           LON           ELEV_FT  ORDER

JT9010   13                        38.348333    -122.302778       62.7
JT9018   13 E                      38.348333    -122.302778       59.0
JT9009   13                        38.346667    -122.301111       58.7
JT9017   13 D                      38.345000    -122.301111       58.7
JT9034   20 A                      38.345000    -122.309444       68.0
JT9012   13                        38.351667    -122.304444       62.5
JT9043   25 A                      38.340000    -122.302778       61.4
JT9016   13 D                      38.341667    -122.297778       59.7
```

CALIFORNIA RESERVOIRS (DWR/CDEC) — within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.94 ft (-9 ft)	706 ft	36K af	**67%**
Folsom Lake	66.5 mi	448.16 ft (-18 ft)	466 ft	787K af	**81%**
Lake Oroville	93.4 mi	861.14 ft (-39 ft)	900 ft	2,858K af	**81%**
New Melones Lake	100.1 mi	1,020.15 ft (-68 ft)	1,088 ft	1,664K af	**69%**
Black Butte Lake	101.2 mi	—	228 ft	74K af	**54%**

USBR RESERVOIRS (Bureau of Reclamation) — within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP

ARMY CORPS RESERVOIRS (USACE) — within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	**100%**
Camanche	89 mi	230.3 ft	221.86 ft	—	**104%**
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	**101%**
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	**107%**
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	**100%**
Don Pedro	137 mi	—	N/R	—	**—**

USGS GROUNDWATER WELLS — within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

⬡ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The California Coastal Ranges Siliciclastic and Terrace Deposits (N100CACSTL) form a heterogeneous aquifer of unconsolidated to semi-consolidated sand, silt, and gravel deposited during Quaternary fluvial and alluvial processes. This geologically young aquifer has moderate to high permeability in sandy intervals but variable yield due to clay lenses and interbedding. Water availability at this Napa County site is moderate, with water table depths ranging from 10 to 88 feet depending on seasonal recharge and local pumping stress.

FORMATION

Name	*California Coastal Ranges Siliciclastic and Terrace Deposits*
Age / Rock	*Quaternary unconsolidated sand, silt, gravel, and clay with local terrace deposits.*
Extent	*800–1200 sq mi (est. USGS/DWR)*

DEPTH & RECHARGE

Depth Range	*51–295 ft (5–30 floors)*
Typical Depth	*100–150 ft (10–15 floors)*
Recharge Source	*Precipitation on valley floor and upland outcrops; lateral flow from Coastal Ranges foothills*
Transit Time	*5–20 years to 100 ft depth*

WATER QUALITY

pH	*est. 6.8–7.5 (est. USGS/DWR)*
Hardness	*est. 150–350 mg/L $CaCO_3$ (est. USGS/DWR)*
TDS	*est. 250–500 mg/L (est. USGS/DWR)*

Notes	*No sample data — typical for formation; minor iron and manganese possible in deeper reducing zones*

TREND

Trend	*Declining from 1977–2014; partial recovery or stabilization by 2022 suggests episodic drought and recharge cycling*
Source	*USGS NWIS historical records and California DWR monitoring*
Seasonal Swing	*est. 8–25 ft seasonal swing; higher fluctuation in shallow wells due to direct recharge response*

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

⬡ AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The Napa Valley region (N100CACSTL aquifer) has not experienced significant land subsidence due to its relatively shallow groundwater table, well-drained soils, and moderate groundwater extraction rates that have not historically caused aquifer compaction. The area's geological setting features primarily unconsolidated to semi-consolidated alluvial and volcanic deposits with good hydraulic conductivity, allowing aquifer recharge and preventing the persistent drawdown conditions that trigger clay layer consolidation. For surveyors working in this area, this means that published benchmarks and elevations are likely stable and reliable for conventional surveying work without subsidence corrections.

SUBSIDENCE ASSESSMENT

Susceptibility	*Low*
Primary Mechanism	*None - consolidated terrain with adequate recharge and moderate withdrawal*
Annual Rate	*<0.5 mm/yr (est.)*
Historic Loss	*None documented*
Aquifer Compaction	*None - aquifer not over-pumped*
SGMA Status	*Not applicable - not in critically overdrafted basin*

BENCHMARK IMPLICATIONS

Monument Risk	*Low - monuments stable*
Elevation Validity	*Good - no active subsidence*
GNSS Recommendation	*Standard benchmark verification adequate*

SURVEY ACTION

Required Action	*No special action required*
Dataset	*CA DWR SGMA Bulletin 118 2024.1*
Coverage	*California critically overdrafted basins only - full US coverage pending*

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	92 mph
Ground Snow Load	5.8 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	**12 in minimum**

Est. from IECC Zone 3C — AFI data unavailable
Verify with local building official before construction

Source: ASCE 7-22 Hazard Tool · gis.asce.org

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

⬡ AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate requires a well-sealed, moderately insulated envelope to manage mild but persistent moisture and occasional fog, with attention to air-sealing details around windows and doors. Orient primary glazing south and east to capture winter sun and passive heating, while minimizing west-facing glass to reduce afternoon heat gain and glare during warmer months. Design operable windows and thermal mass (concrete floors, masonry) to enable natural ventilation and night cooling, which is your most cost-effective strategy for managing the moderate temperature swings between day and night.

⬡ AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal marine site in Wine Country presents well-drained gravelly loam soils suitable for conventional shallow foundations, though seismic parameters are currently unavailable and may warrant confirmation. Moderate wind exposure and light snow loads suggest straightforward structural design, but the Site Class D designation and null seismic coefficients should be resolved early with local building officials and a geotechnical engineer to establish appropriate design criteria.

FOUNDATION

Type	*Conventional spread footing anticipated, subject to geotechnical confirmation*
Basis	*Well-drained Coombs gravelly loam with 2–5% slopes suggests adequate bearing capacity for shallow foundations; consult a licensed geotechnical engineer to confirm bearing pressure and settlement estimates*

STRUCTURAL IMPLICATIONS

Seismic	*Seismic design category and spectral accelerations are unavailable and may warrant clarification with the local building official; consult a structural engineer to establish whether special detailing or ordinary moment-resisting frames are appropriate*
Wind Exposure	*Exposure C (open terrain, Napa/Sonoma coastal area) anticipated at 92 mph 3-second gust per ASCE 7-22 Risk Category II*

Wind	*Standard wind load design likely adequate; no hurricane exposure anticipated for this inland coastal zone*
Snow	*Light-to-moderate snow load of 5.8 psf anticipated; standard roof design likely adequate, though consult engineer for any snow drift or exposure conditions*

INVESTIGATIONS & INSPECTIONS

Geotech Required	*Yes – geotechnical investigation recommended to confirm Site Class D, bearing capacity, settlement, and to verify frost depth (12 in. estimated for IECC planning; local official confirmation suggested)*
Liquefaction Screen	*Screening recommended once seismic parameters are confirmed; well-drained soils suggest low liquefaction risk, but Site Class D and SDC determination should precede detailed evaluation*
Special Inspections	*Scope will depend on final seismic design category; confirm with building official and structural engineer early in design*
Code Edition	*ASCE 7-22 / IBC 2021 (planning reference only)*
Site Class Note	*Site Class D assumed for planning purposes; a Phase I or Phase II geotechnical investigation is suggested to refine soil stratigraphy, groundwater conditions, and confirm design parameters with the local authority having jurisdiction*

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes

SERIES: Coombs

DRAINAGE CLASS: Well drained

HYDRO GROUP: C

SOIL ORDER: Alfisols

SUBORDER: Xeralfs

⬡ AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf suborder) formed in alluvial and colluvial materials on gentle valley footslope and fan positions in the Coast Range and inland valleys of California. The soil profile exhibits a moderately thick surface layer overlying a clay-enriched argillic substratum typical of Alfisols, with significant gravel content throughout that affects both bearing capacity and excavation characteristics. On-site, you will observe a brownish gravelly loam topsoil (10–20 inches) that transitions abruptly to a heavier, more clay-rich substratum; the 2–5% slope gradient generally favors runoff and reduces saturation risk, making this a stable platform for development provided grading accounts for the underlying clay layer.

Alluvial and colluvial parent material on footslopes and fans; native grassland and oak savanna vegetation.

DRAINAGE

Class	*Well drained*
Seasonal Saturation	*None within 60 inches*
Flooding Risk	*Minimal on 2–5% slopes; assess site-specific drainage patterns and proximity to stream channels*
Ponding Risk	*None*

ENGINEERING

Bearing Capacity	*2,500–3,500 psf (shallow foundations); gravel content provides adequate bearing but clay substratum limits deeper loads*
Rating	*Fair*
Shrink-Swell	*Moderate — argillic clay substratum exhibits moderate shrink-swell potential when exposed to wetting-drying cycles*
Frost Action	*Low*
Cut Suitability	*Good — gravel-rich topsoil and upper profile; material is readily excavated and suitable for fill or road base*
Fill Suitability	*Fair — clay substratum requires moisture condition management and compaction control; pre-construction testing recommended*

SEPTIC

Suitability	*Conditional*
Notes	*Perc test required — argillic clay substratum at 20–40 inches depth may restrict effluent percolation; seasonal high-water table and slope position must be verified; consult local health department and conduct detailed site evaluation before system design*

HAZARDS

Excavation	*Moderate — 15–35% gravel content throughout profile; hand-trenching and equipment selection should account for gravelly matrix*
Expansive Clay	*Moderate — clay substratum present; monitor for differential settlement and foundation distress if subgrade is disturbed or saturated*
Erosion	*Low on 2–5% slopes; gully erosion risk increases in disturbed areas and near cut faces; stabilize graded surfaces promptly*
Other	*Verify depth to clay substratum and perform standard geotechnical investigation for structural loads; monitor groundwater response to grading and drainage modifications*

Data sources: California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 13, 2026, 2:41 AM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

🛢 OIL & GAS WELLS California — within 10mi radius

8	0	7
total	active	plugged

By status:

7 Plugged **1** Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	**Plugged**	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	**Idle**	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	**Plugged**	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation dry hole located 7.2 miles away, which is well beyond any actionable review distance. You should confirm whether any surface easements or pipeline rights-of-way cross the property itself, as those could affect site layout regardless of well distance.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

⛏ MINES & MINERAL DEPOSITS USGS MRDS — within ~35mi radius

200 total in search area

10 within 5 mi

108 within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	**Occurrence**	Sand/Gravel	1.6 mi
Basalt Rock Co	**Producer**	Clay	2 mi
Cicero Pumice Quarry	**Occurrence**	PUM	2.9 mi
Mount George King Pumice Deposit.	**Occurrence**	PUM	2.9 mi
Basalt Rock Co.	**Occurrence**	DIT	3.1 mi
Mt George Pumice Deposit	**Producer**	PUM	3.1 mi
Wilson Pumice Quarry	**Occurrence**	PUM	3.7 mi
Unknown	**Occurrence**	Mercury	4.4 mi
Silva Pumice Quarry	**Occurrence**	PUM	4.7 mi
Silva Pumice Quarry	**Producer**	PUM	5 mi
Walker Pumice Quarry	**Occurrence**	PUM	5.1 mi
Walker Pumice Quarry	**Producer**	PUM	5.1 mi
Mountain	**Occurrence**	Mercury	5.4 mi
Unknown	**Occurrence**	Mercury	7.6 mi
Unknown	**Occurrence**	Stone/Crushed	7.8 mi
Unnamed Location	**Past Producer**	Sand/Gravel	8.5 mi
Unnamed Location	**Past Producer**	Sand/Gravel	8.5 mi
Unnamed Prospect	**Occurrence**	Chromium	8.8 mi
Unnamed Location	**Occurrence**	Chromium	9 mi
Unknown	**Occurrence**	Chromium	9.1 mi
Bella Oak	**Past Producer**	Mercury	9.4 mi
Unknown	**Occurrence**	MG	9.7 mi
Serres Pit	**Past Producer**	Sand/Gravel	10.4 mi
Unnamed Location	**Unknown**	Sand/Gravel	10.5 mi
Conn Valley	**Occurrence**	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Three active producers operate within 5 miles of the site. Basalt Rock Co. (2 mi) and Mt George Pumice Deposit (3.1 mi) will generate heavy truck traffic, dust, and potential noise from crushing/blasting operations during active hours. Silva Pumice Quarry (5 mi) presents similar operational impacts. No inactive deposits within 1 mile warrant Phase 1 ESA consideration.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 Endangered

6 Threatened

12 Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	*SCIENTIFIC NAME*	STATUS
California tiger Salamander	*Ambystoma californiense*	**Endangered**

COMMON NAME	*SCIENTIFIC NAME*	STATUS
Conservancy fairy shrimp	*Branchinecta conservatio*	**Endangered**
Contra Costa goldfields	*Lasthenia conjugens*	**Endangered**
Soft bird's-beak	*Cordylanthus mollis ssp. mollis*	**Endangered**
Suisun thistle	*Cirsium hydrophilum var. hydrophilum*	**Endangered**
Vernal pool tadpole shrimp	*Lepidurus packardi*	**Endangered**
Alameda whipsnake (=striped racer)	*Masticophis lateralis euryxanthus*	**Threatened**
California red-legged frog	*Rana draytonii*	**Threatened**
Delta smelt	*Hypomesus transpacificus*	**Threatened**
Northern spotted owl	*Strix occidentalis caurina*	**Threatened**
Vernal pool fairy shrimp	*Branchinecta lynchi*	**Threatened**
western snowy plover	*Charadrius nivosus nivosus*	**Threatened**

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Review Summary – 38.3459, -122.3047 Section 7 consultation with USFWS is triggered if the proposed project involves federal permits, funding, or actions; the site's proximity to critical habitat for 12 species necessitates effects analysis. The most planning-relevant species are California tiger salamander and vernal pool species (tadpole shrimp, fairy shrimp) due to their dependence on seasonal wetlands—common development conflict points—and California red-legged frog and Delta smelt given regional water management implications.

Recommended next steps include: (1) confirming project footprint and federal nexus, (2) conducting focused biological surveys during appropriate seasonal windows to determine actual species presence/absence on-site, and (3) preparing a preliminary effects assessment to scope consultation intensity and permitting timeline. Early coordination with USFWS is advisable to streamline permitting and identify avoidance/mitigation opportunities. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent

COMMON NAME	*SCIENTIFIC NAME*	LAST OBS
American black nightshade	*Solanum americanum*	2026-05-29
Bearded Clover	*Trifolium barbigerum*	2026-05-15
bird's foot cliffbrake	*Pellaea mucronata*	2026-04-28
Blow Wives	*Achyrachaena mollis*	2026-05-20
blue oak	*Quercus douglasii*	2026-05-09
Bolander's Phacelia	*Phacelia bolanderi*	2026-04-27
box elder	*Acer negundo*	2026-05-18
Braun's Giant Horsetail	*Equisetum telmateia braunii*	2026-06-22
California bay	*Umbellularia californica*	2026-06-24
California beeplant	*Scrophularia californica*	2026-05-20
California black oak	*Quercus kelloggii*	2026-05-15
California buckeye	*Aesculus californica*	2026-06-02
California Dutchman's Pipe	*Aristolochia californica*	2026-05-26
California milkwort	*Rhinotropis californica*	2026-05-15
California mugwort	*Artemisia douglasiana*	2026-06-02
California poppy	*Eschscholzia californica*	2026-06-24
California Wild Rose	*Rosa californica*	2026-05-09
Canyon liveforever	*Dudleya cymosa*	2026-05-30
-	*Castilleja densiflora densiflora*	2026-05-09
clay mariposa lily	*Calochortus argillosus*	2026-05-15
coast live oak	*Quercus agrifolia*	2026-05-13
coastal woodfern	*Dryopteris arguta*	2026-05-20
common selfheal	*Prunella vulgaris*	2026-06-22
Diogenes' lantern	*Calochortus amabilis*	2026-05-15
distant phacelia	*Phacelia distans*	2026-05-15
-	*Downingia concolor concolor*	2026-05-15
Dwarf Brodiaea	*Brodiaea terrestris*	2026-05-15
Elegant Clarkia	*Clarkia unguiculata*	2026-05-16
goldback fern	*Pentagramma triangularis*	2026-05-20
Ithuriel's Spear	*Triteleia laxa*	2026-06-15
maroonspot calicoflower	*Downingia concolor*	2026-05-15

COMMON NAME	*SCIENTIFIC NAME*	LAST OBS
narrowleaf milkweed	*Asclepias fascicularis*	2026-06-25
narrowleaf mule-ears	*Wyethia angustifolia*	2026-05-15
Narrowleaf Willow	*Salix exigua*	2026-06-24
needleleaf navarretia	*Navarretia intertexta*	2026-05-15
northern California black walnut	*Juglans hindsii*	2026-06-24
orange bush monkeyflower	*Diplacus aurantiacus*	2026-05-29
Oregon Ash	*Fraxinus latifolia*	2026-06-02
Pacific Bleeding Heart	*Dicentra formosa*	2026-05-04
Pacific madrone	*Arbutus menziesii*	2026-07-04
Pacific poison oak	*Toxicodendron diversilobum*	2026-06-02
pineapple-weed	*Matricaria discoidea*	2026-05-20
Redwood Sorrel	*Oxalis smalliana*	2026-05-28
small-headed clover	*Trifolium microcephalum*	2026-05-15
Superb Mariposa Lily	*Calochortus superbus*	2026-06-11
thimble clover	*Trifolium microdon*	2026-05-15
tomcat clover	*Trifolium willdenovii*	2026-05-15
Toyon	*Heteromeles arbutifolia*	2026-06-19
trailing blackberry	*Rubus ursinus*	2026-06-24
Tree Clover	*Trifolium ciliolatum*	2026-05-15
valley oak	*Quercus lobata*	2026-06-24
wavy-leafed soap plant	*Chlorogalum pomeridianum*	2026-05-20
western sycamore	*Platanus racemosa*	2026-05-17
white brodiaea	*Triteleia hyacinthina*	2026-05-15
white-tipped clover	*Trifolium variegatum*	2026-05-15
Winecup Clarkia	*Clarkia purpurea*	2026-05-15
Yellow Mariposa Lily	*Calochortus luteus*	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 38.3459, -122.3047 This site supports a **mixed oak woodland transitioning to chaparral**, dominated by blue oak and California black oak with California bay in more mesic microsites, characteristic of the inner Coast Ranges. The diverse understory—featuring California buckeye, California poppy, beeplant, and multiple clover species—indicates well-drained, seasonally dry conditions typical of Mediterranean-climate foothill sites with moderate soil fertility. Soil moisture appears **seasonal and limited**, with winter/spring recharge followed by pronounced summer drought; this suggests well-draining soils (likely loamy or decomposed granite) that will shed water readily during grading operations, minimizing ponding but requiring erosion controls on slopes. Drainage design should account for the site's natural tendency toward rapid runoff and shallow groundwater infiltration rather than retention. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants

VENOMOUS SNAKES

SPECIES	OBSERVATIONS
Crotalus oreganus oreganus	635 obs

VENOMOUS SPIDERS

SPECIES	OBSERVATIONS
Latrodectus hesperus	154 obs

STINGING INSECTS

SPECIES	OBSERVATIONS
Dasymutilla aureola	91 obs
Dasymutilla californica	91 obs
Dasymutilla sackenii	91 obs

TOXIC PLANTS

SPECIES	OBSERVATIONS
Toxicodendron diversilobum	1100 obs
Datura wrightii	57 obs
Datura stramonium	57 obs

SPECIES	OBSERVATIONS
Datura ferox	57 obs
Conium maculatum	199 obs
Urtica gracilis	83 obs
Urtica gracilis holosericea	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief: Napa Valley Region Survey Area Your highest-priority hazards are poison oak (Toxicodendron diversilobum)—extremely common in this area with 1,439 observations—and Northern Pacific rattlesnakes, which are abundant with 635 documented sightings. For poison oak, wear long pants, long sleeves, and closed-toe boots; wash clothing separately after work and never burn vegetation. For rattlesnakes, watch where you step and place your hands, stay on cleared paths when possible, and make noise while walking to give snakes time to move away. Black widow spiders and various Datura plants (toxic if ingested) are secondary concerns—avoid dark sheltered spots and never touch unfamiliar plants. Note that snakes are most active April through October and poison oak causes severe skin reactions year-round, so peak caution is needed now through fall. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3 Beam Paths

545ft Closest Beam

3 Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26 Total Airways

13 Jet Routes

2 Victor Airways

11 Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200	90	65	5	14	6
Total Observations	Total Species	Birds	Mammals	Reptiles	Amphibians

BIRDS (65 species)

COMMON NAME	*Scientific Name*	OBS
Wild Turkey	*Meleagris gallopavo*	10
Snowy Egret	*Egretta thula*	7
California Quail	*Callipepla californica*	5
Western Bluebird	*Sialia mexicana*	5
California Towhee	*Melozone crissalis*	5
Great Egret	*Ardea alba*	4
Red-tailed Hawk	*Buteo jamaicensis*	4
Black Phoebe	*Sayornis nigricans*	4
Northern Flicker	*Colaptes auratus*	3
Cooper's Hawk	*Astur cooperii*	3
Ash-throated Flycatcher	*Myiarchus cinerascens*	3
Western Screech-Owl	*Megascops kennicottii*	3
Turkey Vulture	*Cathartes aura*	3
Mallard	*Anas platyrhynchos*	3
Bushtit	*Psaltriparus minimus*	3

+ 50 additional species

MAMMALS (5 species)

COMMON NAME	*Scientific Name*	OBS
Black-tailed Jackrabbit	*Lepus californicus*	3
Coyote	*Canis latrans*	2
Mule Deer	*Odocoileus hemionus*	2
Eastern Fox Squirrel	*Sciurus niger*	1
Striped Skunk	*Mephitis mephitis*	1

REPTILES (14 species)

COMMON NAME	*Scientific Name*	OBS
Western Fence Lizard	*Sceloporus occidentalis*	10
Pacific Gopher Snake	*Pituophis catenifer catenifer*	7
Northern Pacific Rattlesnake	*Crotalus oreganus oreganus*	6
California King Snake	*Lampropeltis californiae*	3
Pond Slider	*Trachemys scripta*	2
Western Skink	*Plestiodon skiltonianus*	2
Ring-necked Snake	*Diadophis punctatus*	1
Common Garter Snake	*Thamnophis sirtalis*	1
Western Yellow-bellied Racer	*Coluber constrictor mormon*	1

COMMON NAME	*Scientific Name*	OBS
Pacific Ringneck Snake	*Diadophis punctatus amabilis*	1
Northwestern Pond Turtle	*Actinemys marmorata*	1
Common Sharp-tailed Snake	*Contia tenuis*	1
North American Racer	*Coluber constrictor*	1
Southern Alligator Lizard	*Elgaria multicarinata*	1

AMPHIBIANS (6 species)

COMMON NAME	*Scientific Name*	OBS
Pacific chorus frog	*Pseudacris regilla*	5
American Bullfrog	*Lithobates catesbeianus*	3
Foothill Yellow-legged Frog	*Rana boylii*	3
Western Toad	*Anaxyrus boreas*	2
Rough-skinned Newt	*Taricha granulosa*	1
California Giant Salamander	*Dicamptodon ensatus*	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in Sonoma County, California supports a diverse wildlife community characteristic of oak woodland-grassland transitional habitat with adjacent riparian or wetland features, as evidenced by the presence of both upland species (California Quail, Western Bluebird, Coyote) and water-dependent species (egrets, pond turtles, chorus frogs). The amphibian assemblage—particularly Foothill Yellow-legged Frog, Rough-skinned Newt, and California Giant Salamander—indicates the presence of cool, flowing or high-quality aquatic habitats that are sensitive to sedimentation, temperature changes, and hydrologic alteration. Several species warrant regulatory attention: Foothill Yellow-legged Frog and Northwestern Pond Turtle are California Species of Special Concern; Northern Pacific Rattlesnake habitat should be avoided where feasible; and riparian areas supporting egrets and salamanders may trigger jurisdictional protections under Fish and Game Code §1602 and the Clean Water Act. The overall species richness and diversity suggests moderate habitat quality with low to moderate existing disturbance, though the presence of non-native American Bullfrog and Eastern Fox Squirrel indicates some anthropogenic influence; development should prioritize riparian buffer maintenance and protection of seasonal breeding ponds. For planning purposes only. Consult a qualified biologist for site-specific biological assessments.

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

☀ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = **38.3°** (latitude) → **9/12 pitch**

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	← excellent
7/12	30.3°	98.8%	← excellent
8/12	33.7°	99.3%	← excellent
9/12	36.9°	99.8%	← annual optimal
10/12	39.8°	99.8%	← excellent
12/12	45°	99%	← excellent

Going from 5/12 to 6/12 at framing costs ~$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~$108 value/yr). Over 25yr panel life: ~$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr·ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)

4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 · THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 · TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	⚠ verify
50ft	All year	4.8	4.6	7.3	⚠ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
⚠ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†Balanced = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
⚠ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

⬡ AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan You're in a sweet spot for solar—you get about 6.6 hours of good sun every single day on average, which is like having a steady paycheck of daylight energy. Your roof pitch is almost perfect for capturing that sun year-round, and here's the thing most people miss: a dark roof will bake your home like an oven in summer (absorbing 301 BTUs per hour per square foot), while a light-colored roof cuts that nearly in half—so choosing roof color is almost as important as insulation. Speaking of heat, that west-facing glass in your home is your biggest problem in summer afternoons, letting in twice as much unwanted heat as your south windows, so you'll want to plan for serious shading there or consider reflective treatments. The smart move is planting a deciduous tree on the south side—it'll give you 4 hours of shade in summer when you need it most, but drop its leaves in winter to let that 51 BTUs of free warmth through your south glass when it's cold. Think of thermal mass (like a concrete floor or adobe wall) as a slow-release heat battery: it soaks up warmth during the day and gently releases it at night, which your light-frame construction does less of than heavier buildings, but it's still worth understanding for your design. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

🧠 **SITE INTELLIGENCE SUMMARY** AI synthesis — Claude Sonnet

⚙ Technical 👤 Client

Technical Brief — for architects, engineers, and surveyors

Site Intelligence Briefing **38.34586°N, 122.30466°W | Napa County, California**

Structural Design Drivers & Climate Envelope

This site falls within IECC Zone 3C (Marine) with an Annual Freezing Index of 0°F-days, indicating a mild marine climate without freeze-thaw cycling as a design constraint. Wind, snow, and seismic data are not available for this location within the standard reference database. Without documented wind speed or snow load parameters, structural design cannot rely on quantified lateral or vertical environmental loads at this stage. The absence of seismic zone classification in the provided dataset is notable; however, Napa County's known seismic activity history warrants independent verification of seismic design category through the latest USGS seismic hazard maps and California Building Code Section 11.4 before framing decisions. The benign freezing index (0°F-days) eliminates frost-driven foundation heave as a design driver; footing depth will be governed instead by bearing capacity and drainage requirements rather than frost penetration.

Geotechnical Profile & Foundation Implications

The site is underlain by Coombs gravelly loam, 2 to 5 percent slopes, classified as Hydrologic Group C with well-drained characteristics. Group C soils indicate moderate infiltration rates and moderate runoff potential, typical of gravelly loam matrices. The well-drained designation is favorable for shallow foundation design, as it reduces hydrostatic pressure risk and minimizes seasonal water table fluctuation at typical footing elevations. Soil texture documentation is incomplete in the provided data; a Phase I geotechnical investigation should confirm fines content, bearing capacity (presumptive or engineered), and actual water table depth, particularly given the 2 to 5 percent slope. The gravelly loam matrix suggests adequate bearing capacity for conventional spread footings or shallow systems, but site-specific CBR or direct shear testing will be required before footing sizing. No frost depth restriction applies; minimum footing depth can be set at 12–18 inches below finish grade or as required by local code, whichever governs.

Subsurface Hazards & Fault Assessment

No Quaternary faults are recorded within the search radius, and no active subsidence basin has been identified for this location. Oil and gas activity is minimal: eight wells exist within a 10-mile radius, but none are active or producing; the nearest well (Mobil Oil Corporation, 7.2 miles distant) is plugged and abandoned. This substantially reduces hydrocarbon-related subsidence risk. However, 200 MRDS mineral deposits lie within approximately 35 miles, and three active mining operations are notably closer: Basalt Rock Co (clay) at 2 miles, Mt George Pumice Deposit at 3.1 miles, and Silva Pumice Quarry at 5 miles. Vibration, dust, and blast effects from nearby quarry operations should be evaluated if sensitive structures (laboratories, precision equipment, or occupied spaces requiring vibration control) are planned. Ground conditions within a 2-mile radius of active extraction warrant inquiry regarding cumulative subsidence or changes in groundwater regime attributable to mining.

Site Context & Regulatory Envelope

The 2-mile proximity to the Basalt Rock Co clay extraction facility places this site in a zone where dust and operational noise may require mitigation design, depending on building use classification and prevailing wind patterns. The marine climate zone (3C) carries no extreme temperature swing requirements, allowing cost-optimized HVAC and envelope design without heavy thermal mass or deep overhangs. The well-drained soil and gentle slope (2–5 percent) are favorable for stormwater management; site drainage toward downslope should be verified during grading design. Napa County's ongoing sensitivity to wildfire and smoke events is not quantified in the provided data but should be cross-referenced with CAL FIRE hazard maps and local emergency management records before occupancy classification decisions are finalized.

Bottom Line: Obtain site-specific seismic design category (California Building Code 11.4 and USGS maps), execute Phase I geotech with water table and bearing capacity confirmation, and verify blast/vibration effects from the 2-mile clay quarry before finalizing structural framing and footing design.

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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

SITE SUMMARY

Coordinates	38.345860°N 122.304660°W
Elevation	66 ft NAVD88 (approx)
Township/Range	T6N, R4W, Mount Diablo Meridian
FEMA Zone	Zone X Panel 06055C0510FF
Nearest BM	JT9010 0.2mi 62.7ftft
Nearest CORS	P261 14.1mi
State Plane	CA 2 (0402)

0 —— 10 —— 20 mi

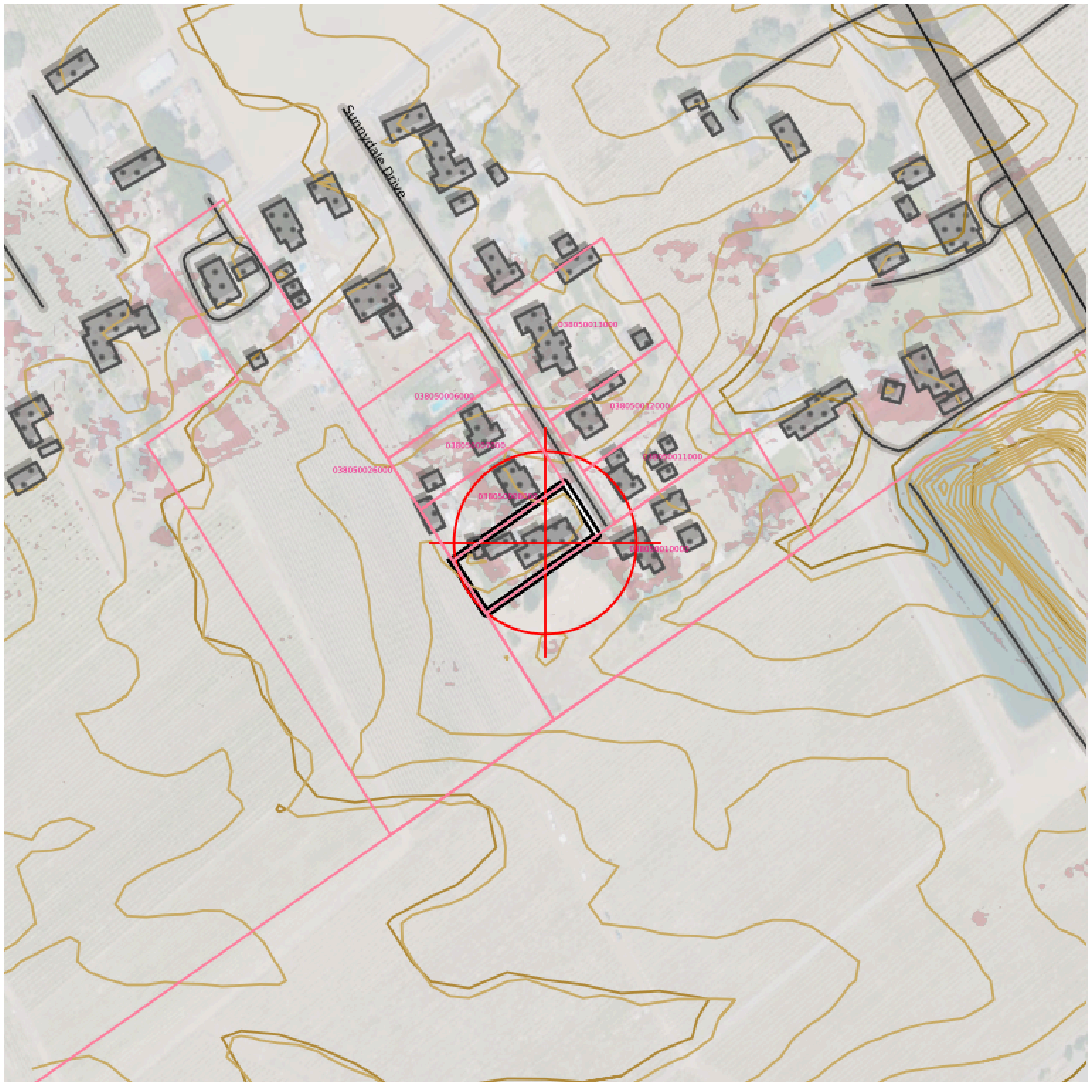
CORS BM SITE FEMA: Zone X Panel 06055C0510FF

TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 13, 2026
Base: USGS National Map
Generated by TopoScout · outKastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | $19 UPGRADE FOR AUTOCAD DXF FILE



N

0 —— 400 ft State Plane · NAD83 · US Feet

INCLUDED LAYERS

■ Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels ■ County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polylines (all geometry)

⚡ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online – https://trimblegnssplanningonline.com/
Satellite visibility, DOP charts, sky plots

FAA B4UFLY – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – https://www.aloft.ai/
Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp – https://www.swpc.noaa.gov/products/planetary-k-index
Real-time Kp index, geomagnetic storm alerts

BASE STATION & NETWORK

NOAA CORS Network – https://geodesy.noaa.gov/CORS/cors-data.shtml
Continuously operating reference stations, data download

NGS CORS Station Map – https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml
Interactive map of all CORS stations with status

IGS Station Monitor – https://www.igs.org/network
International GNSS Service — global base station health

POST-PROCESSING

NGS OPUS – https://geodesy.noaa.gov/OPUS/index.jsp
Online Positioning User Service — static GNSS processing

NGS OPUS-RS – https://geodesy.noaa.gov/OPUSI/index.jsp
Rapid static processing — sessions as short as 15 minutes

CORS Data Download – https://geodesy.noaa.gov/CORS/cors-data.shtml
Raw RINEX observation files from CORS stations

REFERENCE & CALCULATIONS

NGS Datasheets – https://www.ngs.noaa.gov/cgi-bin/datasheet.prl
Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) – https://geodesy.noaa.gov/NCAT/
NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination – https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml
Declination calculator for any location and date

NGS Geoid Models – https://geodesy.noaa.gov/GEOID/
GEOID18 and current model downloads, GEOID calculator

FEMA & FLOOD

FEMA Flood Map Service Center – https://msc.fema.gov/portal/home
FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate
Current EC form, instructions, LOMA/LOMR submission guidance

DRONE OPERATIONS

FAA Part 107 — Commercial Rules – https://www.faa.gov/uas/commercial_operators
Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – https://faadronezone.faa.gov/
Drone registration, operator certificates, waiver management

FAA B4UFLY – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers – https://www.faa.gov/uas/commercial_operators/part_107_waivers
Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – https://www.aloft.ai/
Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

AirNow Fire & Smoke Map
https://www.fire.airnow.gov
Current fire locations, smoke plumes, PM2.5 readings by location

AirNow Interactive Map
https://gispub.epa.gov/airnow
Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

NOAA Smoke Forecast
https://airquality.weather.gov
48-hr smoke concentration forecast, HRRR-Smoke model

Links verified: February 2026 · Check quarterly · TopoScout GNSS Planning System